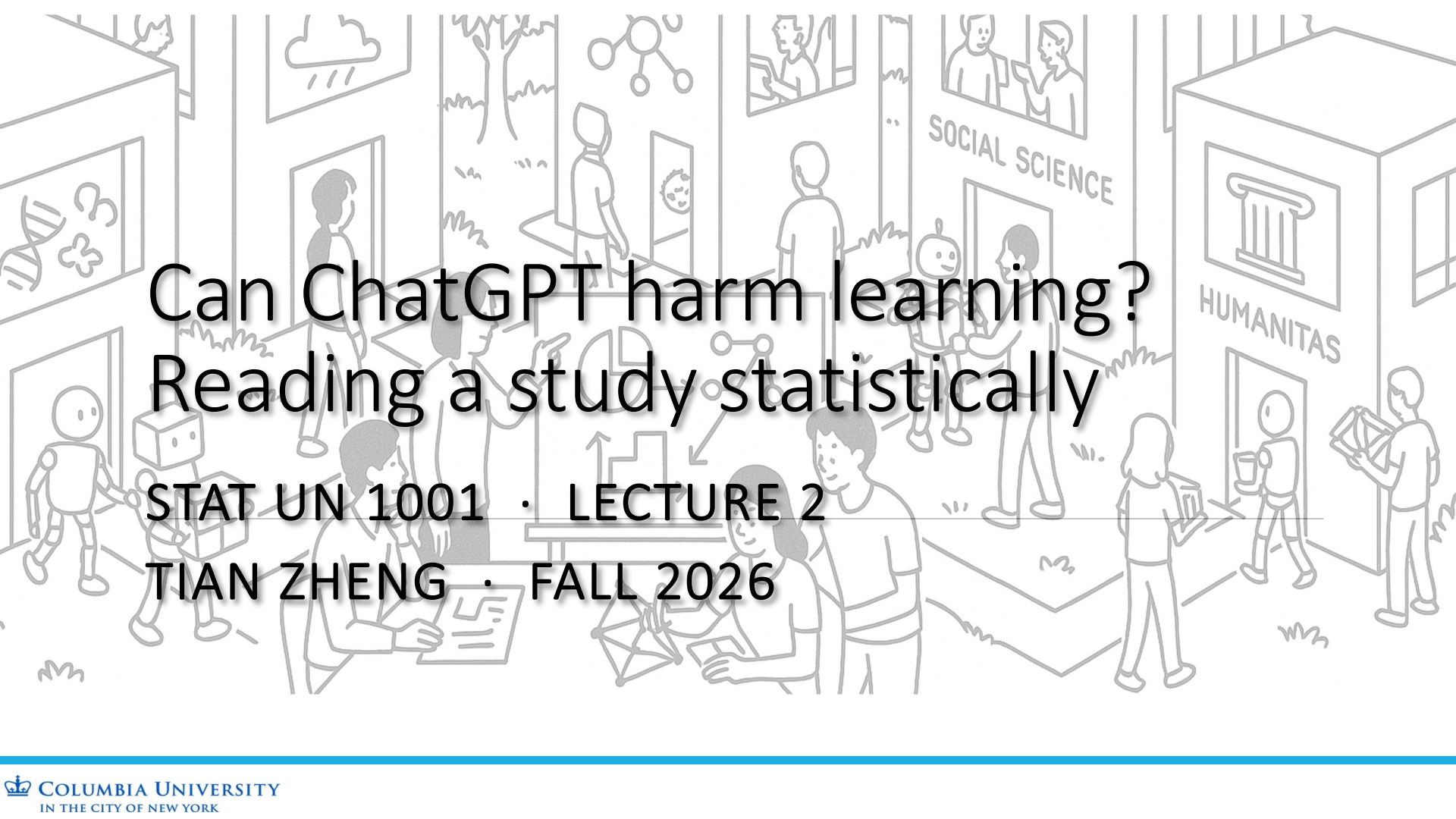




Can ChatGPT harm learning? Reading a study statistically

STAT UN 1001 · LECTURE 2

TIAN ZHENG · FALL 2026

The claim in the news

Using ChatGPT to write essays may be bad for your brain

A 2025 news report said ChatGPT users showed the lowest brain engagement and underperformed on neural, linguistic, and behavioral measures.

What would we need to know before believing that claim?

What does the claim mean?

PAIR DISCUSSION 3 MINUTES

Choose one phrase and make it measurable

“using ChatGPT”

“bad for your brain”

“underperformed”

Write one outcome that researchers could actually record.

Be ready to explain why your outcome represents the idea you chose.

The study question

How does assistance during essay writing relate to neural activity and writing behavior?

The researchers compared ChatGPT, web search, and no external tools during repeated SAT-style essay tasks.

The paper remains a preprint. The authors posted version 1 in June 2025 and revised it in December 2025.

The sample

54

participants in
Sessions 1 to 3

18 to 39

years old

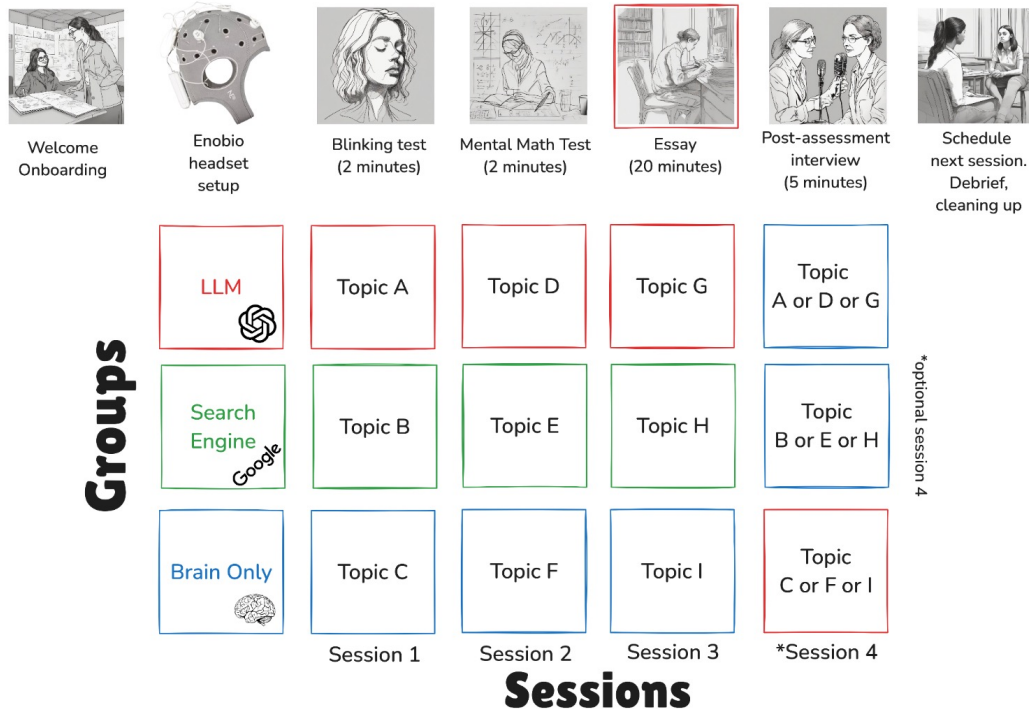
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Greater Boston
universities

The researchers randomly assigned 18 participants to each writing condition, balancing age and gender.

Eighteen participants completed the optional fourth session.

The study design



Sessions 1 to 3

The same assigned tool condition for each participant

Session 4

An optional crossover for the ChatGPT and brain-only groups

Each session included EEG recording, an essay, and a post-task interview.

Figure 5. Study protocol.

Cases, variables, and values

PAIR DISCUSSION 4 MINUTES

Use the handout to complete the study map

Cases

Who or what has one row of data?

Variables

What characteristics did researchers record?

Values

Give one possible value for one variable.

Challenge: Does each participant contribute one observation or several?

The study map

Case

One participant

**Repeated
observation**

One participant in one session

**Explanatory
variable**

Assigned writing condition

**Response
variables**

EEG connectivity, quotation recall, essay features, scores, and reported ownership

A variable names what can vary. A value records what occurred for one observation.

Explanatory and response variables

Writing condition

The assigned condition may explain differences among outcomes.

Measured outcomes

Each outcome captures a different part of writing behavior or neural activity.

Which response variable best represents learning?

The answer depends on the domain question and the meaning of learning.

Random assignment and random sampling

Random assignment

Strengthens comparisons among the three conditions for the participants in this experiment

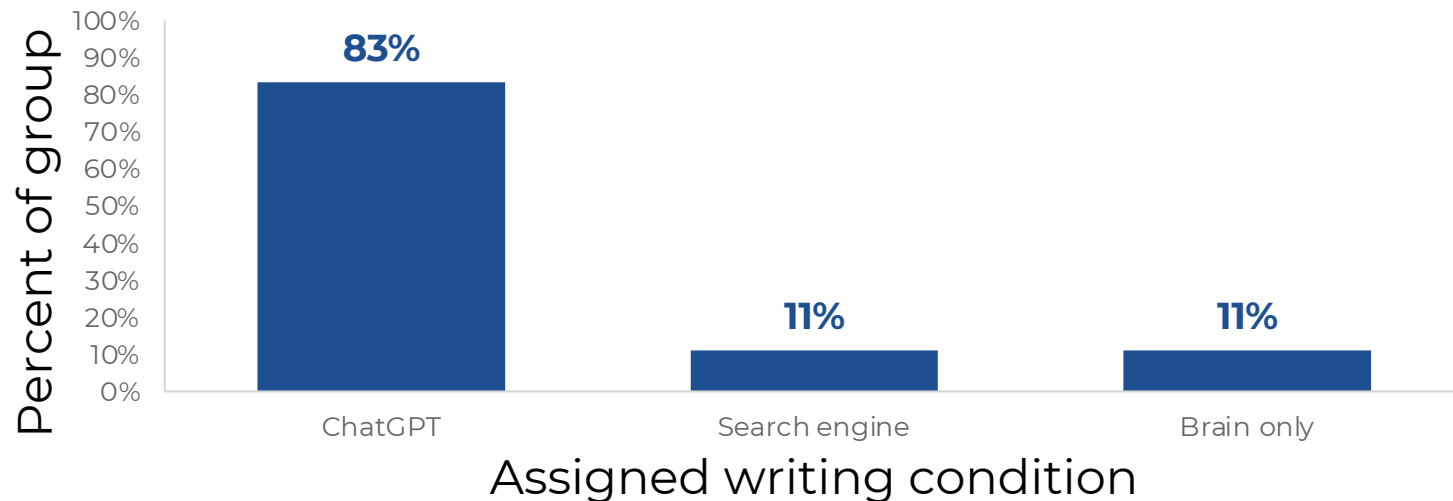
Random sampling

The study did not draw a random sample from all students or all ChatGPT users

The design helps with the comparison inside the experiment. The recruitment limits how far the result can travel.

One reported behavioral result

Participants unable to quote their essay accurately after Session 1



15 of 18 ChatGPT participants, compared with 2 of 18 in each other group

What the authors reported

EEG

The authors described the brain-only group as having the strongest and most distributed connectivity.

Recall

ChatGPT users had more difficulty quoting their own essays in several sessions.

Ownership

Reported ownership tended to be lowest in the ChatGPT group.

These outcomes differ from long-term learning, intelligence, or brain damage.

How the claim grows

PAIR DISCUSSION 4 MINUTES

Observed data

Different
measurements across
assigned groups

Study interpretation

ChatGPT assistance may
reduce engagement
during this task

Public claim

**ChatGPT is bad
for your brain**

**Where does the evidence become too weak
for the claim? Explain one gap.**

What does EEG connectivity represent?

Measured quantity

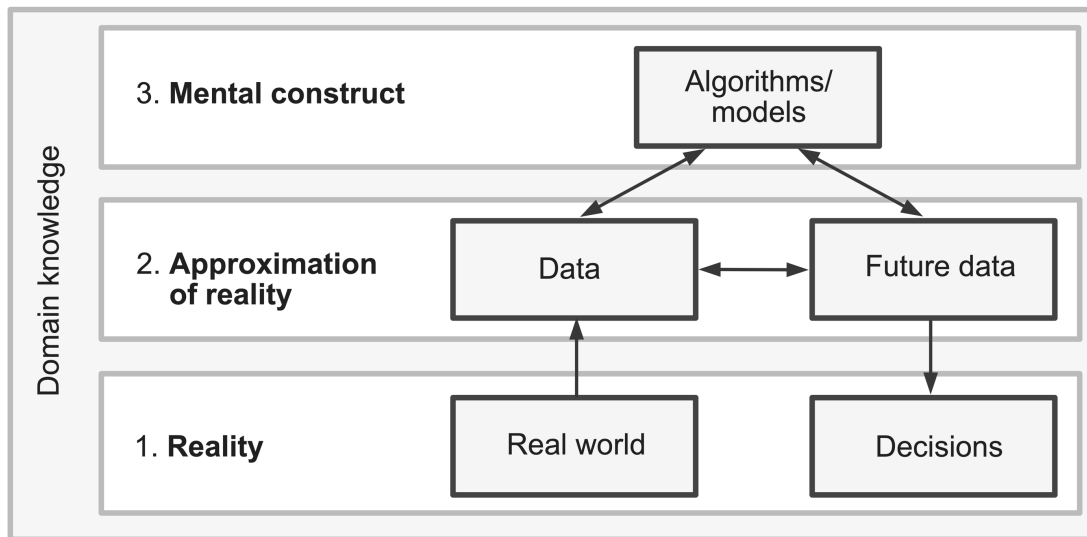
Statistical coupling
among signals recorded
at scalp electrodes
during the task

Target concept

Learning, memory,
critical thinking, or
cognitive engagement

A later methodological comment warned that fewer statistically significant connections do not necessarily mean weaker overall neural activity.

The VDS three-realms check



Reality

How people use AI while learning

Data

EEG signals, essays, and interviews from this experiment

Analysis

Models and judgment calls used to summarize the measurements

Trustworthiness audit

PAIR DISCUSSION 6 MINUTES

Domain question

Does the study ask the question the headline answers?

Data

Do the sample and measurements reflect the relevant reality?

Analysis

Would reasonable analytic choices lead to similar conclusions?

Choose the weakest connection.

A claim the evidence can support

In this sample and constrained essay task, the assigned writing tool was associated with differences in EEG connectivity and several behavioral measures.

Broader conclusions about brain damage, intelligence, or learning across settings require larger and more varied studies with direct learning outcomes.

Good statistical reasoning makes the claim fit the evidence.

Next: Ping Pong AI assistants
